



SaaS Platform

Complete Master Documentation

Zoho Competitor | Full-Stack | Solo Developer Edition

Version 1.0.0 | Date: 2026-05-27 | Author: Mohit Yadav

				~1,09	~1,20		
13	M	98	DB	0	Ci	0+	Files



1. Project Overview

A full-featured multi-tenant SaaS platform competing directly with Zoho. Covers 13 modules: CRM, Forms Builder, Analytics, Project Management, HR, Help Desk, Accounting, Inventory, Workflow Automation, Documents, Team Chat, App Builder, and Core Platform.

Total Modules	13
Database Tables	98
Total Columns (approx)	~1,090
Project Files — Full	~1,200+
Project Files — Phase 1 MVP	~180
API Procedures (tRPC)	~200+
React Components	~300+
Pages / Routes	~60+
Docker Services	5
npm Packages	~45

2. Local Environment Status

Tool	Status	Version / Notes
Node.js	READY	v20.19.0
npm	READY	10.8.2
pnpm	INSTALL NEEDED	npm install -g pnpm
Git	READY	2.51.0
Docker	READY	28.3.3
Docker Compose	READY	v2.39.2
VS Code	READY	Latest
PostgreSQL	VIA DOCKER	16
Redis	VIA DOCKER	7
MinIO	VIA DOCKER	Latest
RAM	SUFFICIENT	13.31 GB
CPU	STRONG	AMD Ryzen 7 7735HS
OS	COMPATIBLE	Windows 11 Enterprise

Pre-requisite commands:

```
npm install -g pnpm
pnpm --version
docker compose up -d
```

3. Complete Tech Stack (2025/2026)

Frontend

Framework	Next.js 15 (App Router + RSC)
Language	TypeScript 5.x
Styling	Tailwind CSS 3.x
UI Components	Shadcn/UI (Radix UI)
Icons	Lucide React
Animations	Framer Motion 11.x
Global State	Zustand 5.x
Server State	TanStack Query v5
Forms	React Hook Form + Zod
Tables	TanStack Table v8
Charts	Recharts + Apache ECharts
Rich Text	Tiptap v2
Drag and Drop	dnd-kit
Dates	date-fns v3
File Upload	React Dropzone
WebSocket Client	Socket.io Client v4

Backend

API Framework	Next.js API Routes + tRPC v11
Type Safety	Zod + TypeScript
ORM	Prisma v6
Job Queues	BullMQ v5 (Redis-based)
File Storage	MinIO (self-hosted S3-compatible)
Email	Nodemailer + React Email

PDF Generation	@react-pdf/renderer
WebSocket Server	Socket.io Server v4
Cron Jobs	node-cron

Database & Cache

Primary Database	PostgreSQL 16 (via Docker)
Cache / Queues	Redis 7 (via Docker)
File Storage	MinIO (via Docker)
Search	PostgreSQL Full-Text Search
ORM Client	Prisma (schema-first)

Authentication

Library	Auth.js v5 (NextAuth)
OAuth Providers	Google, Microsoft, GitHub
MFA / 2FA	TOTP via otplib + Backup codes
Session Store	PostgreSQL (Prisma adapter)
Password Hashing	bcrypt
JWT	jose library

4. Docker Services (Local Infrastructure)

Service	Port	Image	Purpose
PostgreSQL 16	5432	postgres:16-alpine	Primary relational database
Redis 7	6379	redis:7-alpine	Cache + BullMQ job queues
MinIO	9000	minio/minio:latest	S3-compatible file storage
MinIO Console	9001	minio/minio:latest	Web UI for file management
Mailhog	8025	mailhog/mailhog	Email catcher for development

App access points:

Service	URL	Purpose
Next.js Dev Server	http://localhost:3000	Main application
Prisma Studio	http://localhost:5555	Visual DB editor
MinIO Console	http://localhost:9001	File storage UI
Mailhog UI	http://localhost:8025	Dev email viewer

5. Module Summary

#	Module	Tables	Columns	Key Features
1	Core Platform	17	~190	Auth, Multi-tenancy, RBAC, Billing, Audit
2	CRM	12	~135	Contacts, Companies, Leads, Deals, Pipeline
3	Forms Builder	4	~48	Drag-drop builder, Conditional logic, Submissions
4	Analytics & Reports	4	~42	Dashboards, Charts, Report Builder, Schedules
5	Project Management	8	~95	Tasks, Kanban, Gantt, Time Tracking
6	HR & People	8	~105	Employees, Attendance, Leaves, Payroll
7	Help Desk	7	~75	Tickets, SLA, Knowledge Base
8	Workflow Automation	5	~52	Visual Builder, Triggers, App Integrations
9	Accounting	9	~105	Invoices, Expenses, Bank Accounts, Tax
10	Inventory	10	~105	Products, Stock, PO, SO, Warehouses
11	Documents	4	~42	File Storage, Versions, Sharing
12	Team Chat	5	~48	Channels, DMs, Reactions, Threads
13	App Builder	5	~48	Low-code, Custom Tables, Views

#	Module	Tables	Columns	Key Features
	TOTAL	98	~1,090	

6. Build Phases & Timeline

Phase	Module	Timeline	Key Deliverables
1	Foundation + Auth	Weeks 1-4	Monorepo, Docker, Auth.js, Multi-tenancy, Dashboard Shell
2	CRM	Weeks 5-8	Contacts, Companies, Leads, Deals, Pipeline
3	Forms Builder	Weeks 9-12	Drag-drop builder, Submissions, Conditional Logic
4	Analytics & Reports	Weeks 13-16	Dashboards, Charts, Report Builder, Schedules
5	Project Management	Weeks 17-20	Tasks, Board, Gantt, Time Tracking
6	HR & People	Weeks 21-24	Employees, Attendance, Leaves, Payroll
7	Help Desk	Weeks 25-28	Tickets, SLA Policies, Knowledge Base
8	Workflow Automation	Weeks 29-32	Visual Builder, Triggers, Actions, Logs
9	Accounting	Weeks 33-36	Invoices, Expenses, Bank Accounts
10	Inventory	Weeks 37-40	Products, Stock, Purchase/Sales Orders
11	Documents + Chat	Weeks 41-44	File Manager, Team Chat, Reactions
12	App Builder	Weeks 45-48	Low-code, Custom Tables, Views
13	Billing + Launch	Weeks 49-52	Stripe, Admin Panel, Optimization, Launch

7. Environment Variables (.env.local)

Variable	Value / Description
NEXT_PUBLIC_APP_URL	http://localhost:3000
NEXT_PUBLIC_APP_NAME	YourSaaS
NODE_ENV	development
DATABASE_URL	postgresql://postgres:postgres@localhost:5432/saasdb
REDIS_URL	redis://localhost:6379
AUTH_SECRET	min-32-char-random-secret
AUTH_GOOGLE_ID	your-google-client-id
AUTH_GOOGLE_SECRET	your-google-client-secret
AUTH_MICROSOFT_CLIENT_ID	your-microsoft-client-id
AUTH_MICROSOFT_CLIENT_SECRET	your-microsoft-client-secret
AUTH_GITHUB_ID	your-github-client-id
AUTH_GITHUB_SECRET	your-github-client-secret
MINIO_ENDPOINT	localhost
MINIO_PORT	9000
MINIO_ACCESS_KEY	minioadmin
MINIO_SECRET_KEY	minioadmin
MINIO_BUCKET_NAME	saasplatform
SMTP_HOST	localhost (Mailhog)
SMTP_PORT	1025
EMAIL_FROM	noreply@yoursaas.com
STRIPE_SECRET_KEY	sk_test_... (Phase 6)
STRIPE_WEBHOOK_SECRET	whsec_... (Phase 6)
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY	pk_test_... (Phase 6)

8. Complete Database Schema (98 Tables)

All tables use UUID primary keys with `gen_random_uuid()`. All tables include `created_at` `TIMESTAMP DEFAULT NOW()`. Soft-deletes use `deleted_at` `TIMESTAMP DEFAULT NULL`.

MODULE 1 — CORE PLATFORM (Tables 1-17)

TABLE 1: users

Platform user accounts — supports email/password and OAuth users.

Column	Type	Nullable	Default	Description
id	UUID	NO	<code>gen_random_uuid()</code>	Primary key
email	VARCHAR(255)	NO	—	Unique email address
email_verified	BOOLEAN	NO	false	Email verification status
email_verified_at	TIMESTAMP	YES	NULL	When email was verified
name	VARCHAR(255)	NO	—	Full display name
avatar_url	TEXT	YES	NULL	Profile picture URL
password_hash	TEXT	YES	NULL	bcrypt hash (NULL for OAuth)
phone	VARCHAR(20)	YES	NULL	Phone number
phone_verified	BOOLEAN	NO	false	Phone verified flag
timezone	VARCHAR(100)	NO	'UTC'	User timezone
locale	VARCHAR(10)	NO	'en'	Language/locale
is_active	BOOLEAN	NO	true	Account active status
is_superadmin	BOOLEAN	NO	false	Platform superadmin flag
last_login_at	TIMESTAMP	YES	NULL	Last login timestamp
last_login_ip	VARCHAR(45)	YES	NULL	Last login IP
created_at	TIMESTAMP	NO	NOW()	Record creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete timestamp

TABLE 2: organizations

Tenant accounts — each organization is an isolated workspace.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
name	VARCHAR(255)	NO	—	Organization name
slug	VARCHAR(100)	NO	—	UNIQUE URL-friendly identifier
logo_url	TEXT	YES	NULL	Organization logo URL
website	VARCHAR(255)	YES	NULL	Website URL
industry	VARCHAR(100)	YES	NULL	Industry type
size	VARCHAR(50)	YES	NULL	Company size (1-10, 11-50...)
country	VARCHAR(100)	YES	NULL	Country
timezone	VARCHAR(100)	NO	'UTC'	Organization timezone
currency	VARCHAR(10)	NO	'USD'	Default currency
plan_id	UUID	YES	NULL	FK -> plans.id
is_active	BOOLEAN	NO	true	Organization status
trial_ends_at	TIMESTAMP	YES	NULL	Trial period end
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 3: organization_members

Maps users to organizations with roles.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
user_id	UUID	NO	—	FK -> users.id
role_id	UUID	NO	—	FK -> roles.id
is_owner	BOOLEAN	NO	false	Organization owner flag
joined_at	TIMESTAMP	NO	NOW()	Join timestamp
invited_by	UUID	YES	NULL	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 4: roles

Permission roles — can be system-wide or organization-specific.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	YES	NULL	FK -> organizations.id (NULL=system)
name	VARCHAR(100)	NO	—	Role display name
slug	VARCHAR(100)	NO	—	Role identifier
description	TEXT	YES	NULL	Role description
is_system	BOOLEAN	NO	false	System-defined role
is_default	BOOLEAN	NO	false	Default for new members
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 5: permissions

Granular permission definitions per module/action/resource.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
module	VARCHAR(100)	NO	—	Module name (crm, forms, hr...)
action	VARCHAR(100)	NO	—	create read update delete export
resource	VARCHAR(100)	NO	—	Resource (contact, deal, form...)
description	TEXT	YES	NULL	Permission description
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 6: role_permissions

Junction: assigns permissions to roles.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
role_id	UUID	NO	—	FK -> roles.id
permission_id	UUID	NO	—	FK -> permissions.id

Column	Type	Nullable	Default	Description
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 7: sessions

User authentication sessions.

Column	Type	Nullable	Default	Description
id	VARCHAR(255)	NO	—	Session token (PK)
user_id	UUID	NO	—	FK -> users.id
organization_id	UUID	YES	NULL	FK -> organizations.id
expires_at	TIMESTAMP	NO	—	Session expiry
ip_address	VARCHAR(45)	YES	NULL	Client IP
user_agent	TEXT	YES	NULL	Browser info
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 8: oauth_accounts

Linked OAuth provider accounts per user.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
user_id	UUID	NO	—	FK -> users.id
provider	VARCHAR(50)	NO	—	google microsoft github
provider_account_id	VARCHAR(255)	NO	—	Provider user ID
access_token	TEXT	YES	NULL	OAuth access token
refresh_token	TEXT	YES	NULL	OAuth refresh token
expires_at	TIMESTAMP	YES	NULL	Token expiry
token_type	VARCHAR(50)	YES	NULL	Token type (Bearer)
scope	TEXT	YES	NULL	OAuth scopes
id_token	TEXT	YES	NULL	OpenID Connect token
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 9: two_factor_auth

TOTP-based MFA configuration per user.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
user_id	UUID	NO	—	FK -> users.id (UNIQUE)
secret	TEXT	NO	—	Encrypted TOTP secret
backup_codes	TEXT[]	NO	'{}'	Array of hashed backup codes
is_enabled	BOOLEAN	NO	false	2FA enabled status
enabled_at	TIMESTAMP	YES	NULL	When 2FA was enabled
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 10: api_keys

Programmatic API access keys per organization.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
user_id	UUID	NO	—	FK -> users.id
name	VARCHAR(255)	NO	—	Key display name
key_hash	VARCHAR(255)	NO	—	UNIQUE hashed key
key_prefix	VARCHAR(10)	NO	—	First 8 chars for display
scopes	TEXT[]	NO	'{}'	Allowed API scopes
last_used_at	TIMESTAMP	YES	NULL	Last usage time
expires_at	TIMESTAMP	YES	NULL	Expiry (NULL=never)
is_active	BOOLEAN	NO	true	Key active status
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 11: invitations

Pending team member invitations.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
email	VARCHAR(255)	NO	—	Invitee email address
role_id	UUID	NO	—	FK -> roles.id
token	VARCHAR(255)	NO	—	UNIQUE invitation token
invited_by	UUID	NO	—	FK -> users.id
accepted_at	TIMESTAMP	YES	NULL	Acceptance timestamp
expires_at	TIMESTAMP	NO	—	Invitation expiry
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 12: password_reset_tokens

One-time tokens for password recovery.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
user_id	UUID	NO	—	FK -> users.id
token	VARCHAR(255)	NO	—	UNIQUE reset token
expires_at	TIMESTAMP	NO	—	Token expiry
used_at	TIMESTAMP	YES	NULL	When token was used
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 13: audit_logs

Immutable record of all significant platform actions.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	YES	NULL	FK -> organizations.id
user_id	UUID	YES	NULL	FK -> users.id
action	VARCHAR(255)	NO	—	Action name (user.login, etc.)
resource_type	VARCHAR(100)	YES	NULL	Resource type
resource_id	UUID	YES	NULL	Resource UUID

Column	Type	Nullable	Default	Description
old_values	JSONB	YES	NULL	Before-change snapshot
new_values	JSONB	YES	NULL	After-change snapshot
ip_address	VARCHAR(45)	YES	NULL	Client IP
user_agent	TEXT	YES	NULL	Browser info
metadata	JSONB	YES	NULL	Additional context
created_at	TIMESTAMP	NO	NOW()	Event timestamp

TABLE 14: notifications

In-app notifications delivered to users.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
user_id	UUID	NO	—	FK -> users.id
type	VARCHAR(100)	NO	—	Notification type
title	VARCHAR(500)	NO	—	Notification title
message	TEXT	YES	NULL	Body message
data	JSONB	YES	NULL	Extra payload
is_read	BOOLEAN	NO	false	Read status
read_at	TIMESTAMP	YES	NULL	When read
action_url	TEXT	YES	NULL	Click destination URL
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 15: plans

Subscription plan definitions and pricing.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
name	VARCHAR(100)	NO	—	Plan display name
slug	VARCHAR(100)	NO	—	UNIQUE plan identifier
description	TEXT	YES	NULL	Plan description
price_monthly	DECIMAL(10,2)	NO	0.00	Monthly price

Column	Type	Nullable	Default	Description
price_yearly	DECIMAL(10,2)	NO	0.00	Yearly price
currency	VARCHAR(10)	NO	'USD'	Currency code
max_users	INTEGER	YES	NULL	Max users (NULL=unlimited)
max_storage_gb	INTEGER	YES	NULL	Storage limit GB
features	JSONB	NO	'{}'	Feature flags JSON
is_active	BOOLEAN	NO	true	Plan availability
is_free	BOOLEAN	NO	false	Free plan flag
stripe_monthly_price_id	VARCHAR(255)	YES	NULL	Stripe price ID
stripe_yearly_price_id	VARCHAR(255)	YES	NULL	Stripe price ID
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 16: subscriptions

Active billing subscriptions per organization.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
plan_id	UUID	NO	—	FK -> plans.id
status	VARCHAR(50)	NO	'active'	active cancelled past_due trialing
billing_cycle	VARCHAR(20)	NO	'monthly'	monthly or yearly
current_period_start	TIMESTAMP	NO	—	Billing period start
current_period_end	TIMESTAMP	NO	—	Billing period end
cancel_at_period_end	BOOLEAN	NO	false	Cancel at period end
cancelled_at	TIMESTAMP	YES	NULL	Cancellation timestamp
stripe_subscription_id	VARCHAR(255)	YES	NULL	Stripe subscription ID
stripe_customer_id	VARCHAR(255)	YES	NULL	Stripe customer ID
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 17: webhook_endpoints

Outbound webhook destinations per organization.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
url	TEXT	NO	—	Destination URL
description	VARCHAR(500)	YES	NULL	Description
events	TEXT[]	NO	'{}'	Subscribed event types
secret	VARCHAR(255)	NO	—	Signing secret
is_active	BOOLEAN	NO	true	Active status
last_triggered_at	TIMESTAMP	YES	NULL	Last trigger time
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

MODULE 2 — CRM (Tables 18-29)

TABLE 18: crm_pipelines

Sales pipeline definitions.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Pipeline name
description	TEXT	YES	NULL	Description
is_default	BOOLEAN	NO	false	Default pipeline
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 19: crm_pipeline_stages

Stages within a pipeline.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
pipeline_id	UUID	NO	—	FK -> crm_pipelines.id
name	VARCHAR(255)	NO	—	Stage name
order	INTEGER	NO	0	Display order
color	VARCHAR(20)	YES	NULL	Hex color
probability	INTEGER	YES	NULL	Win probability % (0-100)
is_closed_won	BOOLEAN	NO	false	Closed Won stage
is_closed_lost	BOOLEAN	NO	false	Closed Lost stage
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 20: crm_contacts

Individual people in the CRM.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
first_name	VARCHAR(255)	NO	—	First name
last_name	VARCHAR(255)	YES	NULL	Last name
email	VARCHAR(255)	YES	NULL	Email address
phone	VARCHAR(50)	YES	NULL	Phone
mobile	VARCHAR(50)	YES	NULL	Mobile
title	VARCHAR(255)	YES	NULL	Job title
company_id	UUID	YES	NULL	FK -> crm_companies.id
department	VARCHAR(255)	YES	NULL	Department
website	VARCHAR(255)	YES	NULL	Website
linkedin_url	VARCHAR(255)	YES	NULL	LinkedIn URL
twitter_handle	VARCHAR(100)	YES	NULL	Twitter handle
address_line1	VARCHAR(255)	YES	NULL	Address line 1
city	VARCHAR(100)	YES	NULL	City
state	VARCHAR(100)	YES	NULL	State
country	VARCHAR(100)	YES	NULL	Country
postal_code	VARCHAR(20)	YES	NULL	Postal code
avatar_url	TEXT	YES	NULL	Photo URL
source	VARCHAR(100)	YES	NULL	Lead source
status	VARCHAR(50)	NO	'active'	Contact status
lead_score	INTEGER	NO	0	Lead score (0-100)
owner_id	UUID	YES	NULL	FK -> users.id
custom_fields	JSONB	NO	'{}'	Custom field values
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 21: crm_companies

Company/organization records.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Company name
website	VARCHAR(255)	YES	NULL	Website
email	VARCHAR(255)	YES	NULL	Email
phone	VARCHAR(50)	YES	NULL	Phone
industry	VARCHAR(100)	YES	NULL	Industry
size	VARCHAR(50)	YES	NULL	Company size
annual_revenue	DECIMAL(15,2)	YES	NULL	Annual revenue
employees_count	INTEGER	YES	NULL	Employee count
city	VARCHAR(100)	YES	NULL	City
country	VARCHAR(100)	YES	NULL	Country
logo_url	TEXT	YES	NULL	Logo URL
description	TEXT	YES	NULL	Company description
linkedin_url	VARCHAR(255)	YES	NULL	LinkedIn URL
owner_id	UUID	YES	NULL	FK -> users.id
custom_fields	JSONB	NO	'{}'	Custom field values
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 22: crm_leads

Potential customers before qualification.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
first_name	VARCHAR(255)	NO	—	First name

Column	Type	Nullable	Default	Description
last_name	VARCHAR(255)	YES	NULL	Last name
email	VARCHAR(255)	YES	NULL	Email
phone	VARCHAR(50)	YES	NULL	Phone
company_name	VARCHAR(255)	YES	NULL	Company name
title	VARCHAR(255)	YES	NULL	Job title
status	VARCHAR(50)	NO	'new'	new contacted qualified converted lost
source	VARCHAR(100)	YES	NULL	Lead source
lead_score	INTEGER	NO	0	Quality score
notes	TEXT	YES	NULL	Notes
owner_id	UUID	YES	NULL	FK -> users.id
converted_contact_id	UUID	YES	NULL	FK -> crm_contacts.id
converted_deal_id	UUID	YES	NULL	FK -> crm_deals.id
converted_at	TIMESTAMP	YES	NULL	Conversion time
custom_fields	JSONB	NO	'{}'	Custom field values
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 23: crm_deals

Revenue opportunities in the pipeline.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
title	VARCHAR(255)	NO	—	Deal title
value	DECIMAL(15,2)	YES	NULL	Deal value
currency	VARCHAR(10)	NO	'USD'	Currency
pipeline_id	UUID	NO	—	FK -> crm_pipelines.id
stage_id	UUID	NO	—	FK -> crm_pipeline_stages.id
probability	INTEGER	YES	NULL	Win probability %

Column	Type	Nullable	Default	Description
expected_close_date	DATE	YES	NULL	Expected close date
actual_close_date	DATE	YES	NULL	Actual close date
contact_id	UUID	YES	NULL	FK -> crm_contacts.id
company_id	UUID	YES	NULL	FK -> crm_companies.id
owner_id	UUID	YES	NULL	FK -> users.id
status	VARCHAR(50)	NO	'open'	open won lost
lost_reason	TEXT	YES	NULL	Reason for loss
description	TEXT	YES	NULL	Deal description
custom_fields	JSONB	NO	'{}'	Custom field values
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 24: crm_activities

Calls, emails, meetings, tasks for CRM records.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
type	VARCHAR(50)	NO	—	call email meeting task note
subject	VARCHAR(500)	NO	—	Activity subject
description	TEXT	YES	NULL	Details
status	VARCHAR(50)	NO	'pending'	pending completed cancelled
due_date	TIMESTAMP	YES	NULL	Due date/time
completed_at	TIMESTAMP	YES	NULL	Completion time
duration_minutes	INTEGER	YES	NULL	Duration
contact_id	UUID	YES	NULL	FK -> crm_contacts.id
company_id	UUID	YES	NULL	FK -> crm_companies.id
deal_id	UUID	YES	NULL	FK -> crm_deals.id
lead_id	UUID	YES	NULL	FK -> crm_leads.id

Column	Type	Nullable	Default	Description
assigned_to	UUID	YES	NULL	FK -> users.id
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 25: crm_notes

Free-form notes attached to CRM records.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
content	TEXT	NO	—	Note content (rich text)
contact_id	UUID	YES	NULL	FK -> crm_contacts.id
company_id	UUID	YES	NULL	FK -> crm_companies.id
deal_id	UUID	YES	NULL	FK -> crm_deals.id
lead_id	UUID	YES	NULL	FK -> crm_leads.id
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 26: crm_tags

Colour-coded labels for CRM records.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(100)	NO	—	Tag name
color	VARCHAR(20)	NO	'#6366f1'	Hex color
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 27: crm_contact_tags

Junction: contacts <-> tags.

Column	Type	Nullable	Default	Description
contact_id	UUID	NO	—	FK -> crm_contacts.id (PK part)
tag_id	UUID	NO	—	FK -> crm_tags.id (PK part)

TABLE 28: crm_deal_tags

Junction: deals <-> tags.

Column	Type	Nullable	Default	Description
deal_id	UUID	NO	—	FK -> crm_deals.id (PK part)
tag_id	UUID	NO	—	FK -> crm_tags.id (PK part)

TABLE 29: crm_custom_fields

Custom field definitions per CRM module.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
module	VARCHAR(100)	NO	—	contact company deal lead
name	VARCHAR(255)	NO	—	Field display name
slug	VARCHAR(255)	NO	—	Field identifier
type	VARCHAR(50)	NO	—	text number date dropdown checkbox url email
options	JSONB	YES	NULL	Options for dropdown fields
is_required	BOOLEAN	NO	false	Required flag
order	INTEGER	NO	0	Display order
created_at	TIMESTAMP	NO	NOW()	Creation time

MODULE 3 — FORMS BUILDER (Tables 30-33)

TABLE 30: forms

Form definitions and settings.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
title	VARCHAR(500)	NO	—	Form title
description	TEXT	YES	NULL	Description
slug	VARCHAR(255)	NO	—	UNIQUE public URL slug
status	VARCHAR(50)	NO	'draft'	draft published closed
settings	JSONB	NO	'{}'	Notification/redirect settings
theme	JSONB	NO	'{}'	Colors, fonts, branding
is_multi_step	BOOLEAN	NO	false	Multi-step form
success_message	TEXT	YES	'Thank you!'	Post-submit message
redirect_url	TEXT	YES	NULL	Post-submit redirect URL
max_submissions	INTEGER	YES	NULL	Response limit
close_at	TIMESTAMP	YES	NULL	Auto-close date
submission_count	INTEGER	NO	0	Total submissions
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 31: form_fields

Individual fields within a form.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
form_id	UUID	NO	—	FK -> forms.id
step	INTEGER	NO	1	Step number for multi-step

Column	Type	Nullable	Default	Description
type	VARCHAR(50)	NO	—	text email phone number date time datetime dropdown checkbox radio file textarea heading paragraph divider rating signature address
label	VARCHAR(500)	NO	—	Field label
placeholder	VARCHAR(500)	YES	NULL	Placeholder text
help_text	TEXT	YES	NULL	Help/hint text
is_required	BOOLEAN	NO	false	Required field
order	INTEGER	NO	0	Display order
options	JSONB	YES	NULL	Options for select fields
validation	JSONB	YES	NULL	Validation rules
logic	JSONB	YES	NULL	Conditional show/hide rules
settings	JSONB	NO	'{}'	Width, column span etc.
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 32: form_submissions

Individual form submission records.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
form_id	UUID	NO	—	FK -> forms.id
submission_number	INTEGER	NO	—	Sequential number
status	VARCHAR(50)	NO	'new'	new reviewed starred trashed
ip_address	VARCHAR(45)	YES	NULL	Submitter IP
user_agent	TEXT	YES	NULL	Browser info
referrer	TEXT	YES	NULL	Referrer URL
submitted_by	UUID	YES	NULL	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Submission time

TABLE 33: form_submission_data

Field-level values for each submission.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
submission_id	UUID	NO	—	FK -> form_submissions.id
field_id	UUID	NO	—	FK -> form_fields.id
value	TEXT	YES	NULL	Field text value
file_url	TEXT	YES	NULL	File URL (file fields)
created_at	TIMESTAMP	NO	NOW()	Creation time

MODULE 4 — ANALYTICS & REPORTS (Tables 34-37)

TABLE 34: dashboards

Dashboard containers holding multiple widgets.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Dashboard name
description	TEXT	YES	NULL	Description
is_default	BOOLEAN	NO	false	Default dashboard
is_public	BOOLEAN	NO	false	Public access
layout	JSONB	NO	'[]'	Widget grid positions
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 35: dashboard_widgets

Individual chart/KPI widgets on dashboards.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
dashboard_id	UUID	NO	—	FK -> dashboards.id
type	VARCHAR(100)	NO	—	bar_chart line_chart pie_chart donut_chart area_chart kpi_card table funnel scatter
title	VARCHAR(255)	NO	—	Widget title
config	JSONB	NO	'{}'	Query, filters, styling config
position_x	INTEGER	NO	0	Grid column position
position_y	INTEGER	NO	0	Grid row position
width	INTEGER	NO	4	Column span (max 12)
height	INTEGER	NO	3	Row span
created_at	TIMESTAMP	NO	NOW()	Creation time

Column	Type	Nullable	Default	Description
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 36: reports

Saved report configurations.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Report name
description	TEXT	YES	NULL	Description
module	VARCHAR(100)	NO	—	crm hr projects helpdesk accounting
type	VARCHAR(50)	NO	—	tabular summary matrix pivot
config	JSONB	NO	'{}'	Columns, filters, grouping, sorting
is_public	BOOLEAN	NO	false	Public access
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 37: report_schedules

Automated report email schedules.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
report_id	UUID	NO	—	FK -> reports.id
frequency	VARCHAR(50)	NO	—	daily weekly monthly
day_of_week	INTEGER	YES	NULL	0=Sun to 6=Sat (weekly)
day_of_month	INTEGER	YES	NULL	1-31 (monthly)
time_hour	INTEGER	NO	9	Hour (0-23)
time_minute	INTEGER	NO	0	Minute (0-59)
timezone	VARCHAR(100)	NO	'UTC'	Schedule timezone
recipients	TEXT[]	NO	—	Email recipients array

Column	Type	Nullable	Default	Description
format	VARCHAR(20)	NO	'pdf'	pdf xlsx csv
is_active	BOOLEAN	NO	true	Active status
last_sent_at	TIMESTAMP	YES	NULL	Last send time
created_at	TIMESTAMP	NO	NOW()	Creation time

MODULE 5 — PROJECT MANAGEMENT (Tables 38-45)

TABLE 38: projects

Top-level project records.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Project name
description	TEXT	YES	NULL	Description
status	VARCHAR(50)	NO	'active'	active on_hold completed cancelled
priority	VARCHAR(20)	NO	'medium'	low medium high urgent
start_date	DATE	YES	NULL	Start date
end_date	DATE	YES	NULL	End date
budget	DECIMAL(15,2)	YES	NULL	Budget
progress	INTEGER	NO	0	Completion % (0-100)
color	VARCHAR(20)	NO	'#6366f1'	Project color
owner_id	UUID	NO	—	FK -> users.id
client_name	VARCHAR(255)	YES	NULL	Client name
tags	TEXT[]	NO	'{}'	Project tags
custom_fields	JSONB	NO	'{}'	Custom fields
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 39: project_members

Team members assigned to a project.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
project_id	UUID	NO	—	FK -> projects.id
user_id	UUID	NO	—	FK -> users.id

Column	Type	Nullable	Default	Description
role	VARCHAR(50)	NO	'member'	owner manager member viewer
joined_at	TIMESTAMP	NO	NOW()	Join timestamp

TABLE 40: task_lists

Sections/groups within a project.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
project_id	UUID	NO	—	FK -> projects.id
name	VARCHAR(255)	NO	—	Section name
order	INTEGER	NO	0	Display order
color	VARCHAR(20)	YES	NULL	Section color
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 41: tasks

Individual work items (supports subtasks).

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
project_id	UUID	YES	NULL	FK -> projects.id
task_list_id	UUID	YES	NULL	FK -> task_lists.id
parent_task_id	UUID	YES	NULL	FK -> tasks.id (subtask)
title	VARCHAR(500)	NO	—	Task title
description	TEXT	YES	NULL	Rich text description
status	VARCHAR(50)	NO	'todo'	todo in_progress review done
priority	VARCHAR(20)	NO	'medium'	low medium high urgent
assignee_id	UUID	YES	NULL	FK -> users.id
reporter_id	UUID	NO	—	FK -> users.id
start_date	DATE	YES	NULL	Start date
due_date	DATE	YES	NULL	Due date

Column	Type	Nullable	Default	Description
completed_at	TIMESTAMP	YES	NULL	Completion time
estimated_hours	DECIMAL(6,2)	YES	NULL	Estimated hours
actual_hours	DECIMAL(6,2)	NO	0	Logged hours
order	INTEGER	NO	0	Sort order
tags	TEXT[]	NO	'{}'	Tags
custom_fields	JSONB	NO	'{}'	Custom fields
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 42: task_comments

Comments and replies on tasks.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
task_id	UUID	NO	—	FK -> tasks.id
user_id	UUID	NO	—	FK -> users.id
content	TEXT	NO	—	Comment text
is_edited	BOOLEAN	NO	false	Edited flag
parent_id	UUID	YES	NULL	FK -> task_comments.id (reply)
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 43: task_attachments

Files attached to tasks.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
task_id	UUID	NO	—	FK -> tasks.id
name	VARCHAR(255)	NO	—	File name
url	TEXT	NO	—	MinIO storage URL

Column	Type	Nullable	Default	Description
size	BIGINT	YES	NULL	File size bytes
mime_type	VARCHAR(100)	YES	NULL	MIME type
uploaded_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 44: milestones

Project milestone markers.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
project_id	UUID	NO	—	FK -> projects.id
name	VARCHAR(255)	NO	—	Milestone name
description	TEXT	YES	NULL	Description
due_date	DATE	YES	NULL	Target date
status	VARCHAR(50)	NO	'pending'	pending completed
completed_at	TIMESTAMP	YES	NULL	Completion time
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 45: time_logs

Work time entries per task/project.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
task_id	UUID	YES	NULL	FK -> tasks.id
project_id	UUID	YES	NULL	FK -> projects.id
user_id	UUID	NO	—	FK -> users.id
description	TEXT	YES	NULL	Work description
hours	DECIMAL(6,2)	NO	—	Hours logged
log_date	DATE	NO	—	Date of work
billable	BOOLEAN	NO	true	Billable flag

Column	Type	Nullable	Default	Description
hourly_rate	DECIMAL(10,2)	YES	NULL	Override rate
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

MODULE 6 — HR & PEOPLE (Tables 46-53)

TABLE 46: departments

Organizational department hierarchy.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Department name
code	VARCHAR(50)	YES	NULL	Department code
parent_id	UUID	YES	NULL	FK -> departments.id
manager_id	UUID	YES	NULL	FK -> employees.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 47: designations

Job titles/designations.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Designation name
level	INTEGER	YES	NULL	Seniority level
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 48: employees

Employee master records.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
user_id	UUID	YES	NULL	FK -> users.id (linked account)
employee_code	VARCHAR(100)	YES	NULL	Unique employee ID
first_name	VARCHAR(255)	NO	—	First name

Column	Type	Nullable	Default	Description
last_name	VARCHAR(255)	YES	NULL	Last name
email	VARCHAR(255)	NO	—	Work email
phone	VARCHAR(50)	YES	NULL	Phone
department_id	UUID	YES	NULL	FK -> departments.id
designation_id	UUID	YES	NULL	FK -> designations.id
manager_id	UUID	YES	NULL	FK -> employees.id
employment_type	VARCHAR(50)	NO	'full_time'	full_time part_time contract intern
status	VARCHAR(50)	NO	'active'	active inactive terminated
date_of_birth	DATE	YES	NULL	Date of birth
gender	VARCHAR(20)	YES	NULL	Gender
join_date	DATE	NO	—	Employment start date
termination_date	DATE	YES	NULL	End date
pan_number	VARCHAR(50)	YES	NULL	PAN/Tax ID
bank_account_number	VARCHAR(100)	YES	NULL	Bank account
bank_name	VARCHAR(255)	YES	NULL	Bank name
custom_fields	JSONB	NO	'{}'	Custom fields
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 49: leave_types

Leave type definitions (CL, SL, EL, etc.).

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Leave type name
code	VARCHAR(50)	NO	—	Code (CL, SL, EL, PL)
color	VARCHAR(20)	NO	'#10b981'	Display color
days_per_year	DECIMAL(5,1)	NO	—	Days allocated/year
is_paid	BOOLEAN	NO	true	Paid leave flag

Column	Type	Nullable	Default	Description
carry_forward	BOOLEAN	NO	false	Allow carry forward
max_carry_forward_days	DECIMAL(5,1)	YES	NULL	Max carry forward
requires_approval	BOOLEAN	NO	true	Needs manager approval
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 50: leave_requests

Employee leave application records.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
employee_id	UUID	NO	—	FK -> employees.id
leave_type_id	UUID	NO	—	FK -> leave_types.id
start_date	DATE	NO	—	Leave start date
end_date	DATE	NO	—	Leave end date
total_days	DECIMAL(5,1)	NO	—	Number of days
reason	TEXT	YES	NULL	Leave reason
status	VARCHAR(50)	NO	'pending'	pending approved rejected cancelled
approved_by	UUID	YES	NULL	FK -> users.id
approved_at	TIMESTAMP	YES	NULL	Approval time
rejection_reason	TEXT	YES	NULL	Rejection reason
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 51: leave_balances

Per-employee leave quota tracking.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
employee_id	UUID	NO	—	FK -> employees.id

Column	Type	Nullable	Default	Description
leave_type_id	UUID	NO	—	FK -> leave_types.id
year	INTEGER	NO	—	Calendar year
total_days	DECIMAL(5,1)	NO	—	Total allocated
used_days	DECIMAL(5,1)	NO	0	Used days
pending_days	DECIMAL(5,1)	NO	0	Pending approval
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 52: attendance_records

Daily attendance check-in/check-out.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
employee_id	UUID	NO	—	FK -> employees.id
date	DATE	NO	—	Attendance date
check_in	TIMESTAMP	YES	NULL	Check-in time
check_out	TIMESTAMP	YES	NULL	Check-out time
check_in_location	VARCHAR(255)	YES	NULL	Check-in location
check_out_location	VARCHAR(255)	YES	NULL	Check-out location
status	VARCHAR(50)	NO	'present'	present absent late half_day on_leave holiday
total_hours	DECIMAL(5,2)	YES	NULL	Total work hours
overtime_hours	DECIMAL(5,2)	NO	0	Overtime hours
notes	TEXT	YES	NULL	Notes
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 53: payroll_records

Monthly payroll calculations.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
employee_id	UUID	NO	—	FK -> employees.id
month	INTEGER	NO	—	Month (1-12)
year	INTEGER	NO	—	Year
basic_salary	DECIMAL(12,2)	NO	—	Basic salary
allowances	JSONB	NO	'{}'	Allowance breakdown
deductions	JSONB	NO	'{}'	Deduction breakdown
gross_salary	DECIMAL(12,2)	NO	—	Gross salary
net_salary	DECIMAL(12,2)	NO	—	Net take-home
tax_amount	DECIMAL(12,2)	NO	0	Income tax
pf_amount	DECIMAL(12,2)	NO	0	Provident fund
esi_amount	DECIMAL(12,2)	NO	0	ESI contribution
status	VARCHAR(50)	NO	'draft'	draft approved paid
payment_date	DATE	YES	NULL	Payment date
payment_method	VARCHAR(50)	YES	NULL	bank_transfer cash cheque
notes	TEXT	YES	NULL	Notes
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

MODULE 7 — HELP DESK (Tables 54-60)

TABLE 54: ticket_categories

Hierarchical ticket category tree.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Category name
description	TEXT	YES	NULL	Description
parent_id	UUID	YES	NULL	FK -> ticket_categories.id
color	VARCHAR(20)	YES	NULL	Color
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 55: sla_policies

Service level agreement definitions.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Policy name
priority	VARCHAR(20)	NO	—	low medium high urgent
first_response_hours	INTEGER	NO	—	First response limit (hours)
resolution_hours	INTEGER	NO	—	Resolution limit (hours)
business_hours_only	BOOLEAN	NO	true	Business hours only
is_active	BOOLEAN	NO	true	Active status
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 56: tickets

Customer support ticket records.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id

Column	Type	Nullable	Default	Description
ticket_number	VARCHAR(50)	NO	—	Auto-generated number
subject	VARCHAR(500)	NO	—	Ticket subject
description	TEXT	YES	NULL	Issue description
status	VARCHAR(50)	NO	'open'	open pending resolved closed
priority	VARCHAR(20)	NO	'medium'	low medium high urgent
category_id	UUID	YES	NULL	FK -> ticket_categories.id
sla_policy_id	UUID	YES	NULL	FK -> sla_policies.id
customer_name	VARCHAR(255)	YES	NULL	Customer name
customer_email	VARCHAR(255)	YES	NULL	Customer email
assignee_id	UUID	YES	NULL	FK -> users.id
source	VARCHAR(50)	NO	'web'	web email phone chat
first_response_at	TIMESTAMP	YES	NULL	First response time
resolved_at	TIMESTAMP	YES	NULL	Resolution time
due_at	TIMESTAMP	YES	NULL	SLA due date
tags	TEXT[]	NO	'{}'	Tags
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 57: ticket_comments

Agent/customer replies on tickets.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
ticket_id	UUID	NO	—	FK -> tickets.id
user_id	UUID	YES	NULL	FK -> users.id
content	TEXT	NO	—	Comment text
is_private	BOOLEAN	NO	false	Internal note flag
is_customer	BOOLEAN	NO	false	Customer reply flag
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

Column	Type	Nullable	Default	Description
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 58: ticket_attachments

Files attached to tickets or comments.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
ticket_id	UUID	NO	—	FK -> tickets.id
comment_id	UUID	YES	NULL	FK -> ticket_comments.id
name	VARCHAR(255)	NO	—	File name
url	TEXT	NO	—	File URL
size	BIGINT	YES	NULL	File size bytes
mime_type	VARCHAR(100)	YES	NULL	MIME type
uploaded_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 59: knowledge_base_categories

KB article categories.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Category name
parent_id	UUID	YES	NULL	FK -> kb_categories.id
order	INTEGER	NO	0	Display order
icon	VARCHAR(50)	YES	NULL	Icon name
is_public	BOOLEAN	NO	true	Publicly accessible
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 60: knowledge_base_articles

Help articles and documentation.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
category_id	UUID	NO	—	FK -> kb_categories.id
title	VARCHAR(500)	NO	—	Article title
slug	VARCHAR(500)	NO	—	URL slug
content	TEXT	NO	—	Rich text content
status	VARCHAR(50)	NO	'draft'	draft published
is_public	BOOLEAN	NO	true	Public visibility
views	INTEGER	NO	0	View count
helpful_count	INTEGER	NO	0	Helpful votes
not_helpful_count	INTEGER	NO	0	Not helpful votes
tags	TEXT[]	NO	'{'	Tags
author_id	UUID	NO	—	FK -> users.id
published_at	TIMESTAMP	YES	NULL	Publish time
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

MODULE 8 — WORKFLOW AUTOMATION (Tables 61-65)

TABLE 61: workflows

Workflow automation definitions.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Workflow name
description	TEXT	YES	NULL	Description
status	VARCHAR(50)	NO	'draft'	draft active paused
trigger_type	VARCHAR(100)	NO	—	form_submitted record_created schedule webhook manual
trigger_config	JSONB	NO	'{}'	Trigger configuration
created_by	UUID	NO	—	FK -> users.id
run_count	INTEGER	NO	0	Total executions
last_run_at	TIMESTAMP	YES	NULL	Last run time
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 62: workflow_steps

Individual steps within a workflow.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
workflow_id	UUID	NO	—	FK -> workflows.id
type	VARCHAR(100)	NO	—	action condition delay loop
name	VARCHAR(255)	NO	—	Step name
config	JSONB	NO	'{}'	Step configuration
order	INTEGER	NO	—	Execution order
parent_step_id	UUID	YES	NULL	FK -> workflow_steps.id
branch	VARCHAR(20)	YES	NULL	'true' or 'false' branch
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 63: workflow_runs

Execution instances of workflows.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
workflow_id	UUID	NO	—	FK -> workflows.id
status	VARCHAR(50)	NO	'running'	running completed failed cancelled
trigger_data	JSONB	YES	NULL	Data that triggered run
context	JSONB	NO	'{}'	Execution variables
started_at	TIMESTAMP	NO	NOW()	Start time
completed_at	TIMESTAMP	YES	NULL	End time
error_message	TEXT	YES	NULL	Error details

TABLE 64: workflow_run_logs

Per-step execution logs.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
run_id	UUID	NO	—	FK -> workflow_runs.id
step_id	UUID	YES	NULL	FK -> workflow_steps.id
status	VARCHAR(50)	NO	—	success failed skipped
input	JSONB	YES	NULL	Step input
output	JSONB	YES	NULL	Step output
error	TEXT	YES	NULL	Error message
duration_ms	INTEGER	YES	NULL	Execution ms
created_at	TIMESTAMP	NO	NOW()	Log time

TABLE 65: app_integrations

Connected third-party app credentials.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id

Column	Type	Nullable	Default	Description
app_name	VARCHAR(100)	NO	—	App ID (slack, gmail...)
display_name	VARCHAR(255)	NO	—	Display name
credentials	TEXT	NO	—	Encrypted credentials JSON
status	VARCHAR(50)	NO	'active'	active expired disconnected
connected_by	UUID	NO	—	FK -> users.id
connected_at	TIMESTAMP	NO	NOW()	Connection time
expires_at	TIMESTAMP	YES	NULL	Credential expiry
created_at	TIMESTAMP	NO	NOW()	Creation time

MODULE 9 — ACCOUNTING (Tables 66-74)

TABLE 66: acc_chart_of_accounts

Chart of accounts for bookkeeping.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
code	VARCHAR(50)	NO	—	Account code
name	VARCHAR(255)	NO	—	Account name
type	VARCHAR(50)	NO	—	asset liability equity income expense
parent_id	UUID	YES	NULL	FK -> acc_chart_of_accounts.id
description	TEXT	YES	NULL	Description
is_system	BOOLEAN	NO	false	System account
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 67: acc_customers

Billing customers.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
contact_id	UUID	YES	NULL	FK -> crm_contacts.id
name	VARCHAR(255)	NO	—	Customer name
email	VARCHAR(255)	YES	NULL	Email
tax_number	VARCHAR(100)	YES	NULL	GST/VAT number
currency	VARCHAR(10)	NO	'USD'	Currency
payment_terms	INTEGER	NO	30	Terms (days)
credit_limit	DECIMAL(15,2)	YES	NULL	Credit limit
outstanding_balance	DECIMAL(15,2)	NO	0	Outstanding amount
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 68: acc_vendors

Purchase vendors/suppliers.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Vendor name
email	VARCHAR(255)	YES	NULL	Email
phone	VARCHAR(50)	YES	NULL	Phone
tax_number	VARCHAR(100)	YES	NULL	Tax number
currency	VARCHAR(10)	NO	'USD'	Currency
payment_terms	INTEGER	NO	30	Terms (days)
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 69: acc_invoices

Customer invoices.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
invoice_number	VARCHAR(100)	NO	—	Auto-generated number
customer_id	UUID	NO	—	FK -> acc_customers.id
status	VARCHAR(50)	NO	'draft'	draft sent paid overdue cancelled
issue_date	DATE	NO	—	Issue date
due_date	DATE	NO	—	Payment due date
currency	VARCHAR(10)	NO	'USD'	Currency
subtotal	DECIMAL(15,2)	NO	0	Subtotal
tax_amount	DECIMAL(15,2)	NO	0	Tax
discount_amount	DECIMAL(15,2)	NO	0	Discount
total_amount	DECIMAL(15,2)	NO	0	Total
amount_paid	DECIMAL(15,2)	NO	0	Amount received
notes	TEXT	YES	NULL	Notes

Column	Type	Nullable	Default	Description
sent_at	TIMESTAMP	YES	NULL	Sent time
paid_at	TIMESTAMP	YES	NULL	Payment time
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 70: acc_invoice_items

Line items within an invoice.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
invoice_id	UUID	NO	—	FK -> acc_invoices.id
description	VARCHAR(500)	NO	—	Item description
quantity	DECIMAL(10,3)	NO	1	Quantity
unit_price	DECIMAL(15,2)	NO	—	Price per unit
discount_percent	DECIMAL(5,2)	NO	0	Discount %
tax_rate	DECIMAL(5,2)	NO	0	Tax rate %
amount	DECIMAL(15,2)	NO	—	Line total
order	INTEGER	NO	0	Display order
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 71: acc_expenses

Expense records.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
category	VARCHAR(255)	NO	—	Expense category
amount	DECIMAL(15,2)	NO	—	Amount
currency	VARCHAR(10)	NO	'USD'	Currency
vendor_id	UUID	YES	NULL	FK -> acc_vendors.id
description	TEXT	YES	NULL	Description

Column	Type	Nullable	Default	Description
expense_date	DATE	NO	—	Date
receipt_url	TEXT	YES	NULL	Receipt file URL
status	VARCHAR(50)	NO	'pending'	pending approved rejected paid
submitted_by	UUID	NO	—	FK -> users.id
approved_by	UUID	YES	NULL	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 72: acc_bank_accounts

Bank and cash accounts.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Account name
account_number	VARCHAR(100)	YES	NULL	Account number
bank_name	VARCHAR(255)	YES	NULL	Bank name
account_type	VARCHAR(50)	NO	—	checking savings credit_card cash
currency	VARCHAR(10)	NO	'USD'	Currency
opening_balance	DECIMAL(15,2)	NO	0	Opening balance
current_balance	DECIMAL(15,2)	NO	0	Current balance
is_primary	BOOLEAN	NO	false	Primary account
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 73: acc_transactions

Financial transaction ledger.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
bank_account_id	UUID	YES	NULL	FK -> acc_bank_accounts.id

Column	Type	Nullable	Default	Description
type	VARCHAR(50)	NO	—	income expense transfer
amount	DECIMAL(15,2)	NO	—	Amount
currency	VARCHAR(10)	NO	'USD'	Currency
date	DATE	NO	—	Transaction date
description	TEXT	YES	NULL	Description
reference	VARCHAR(255)	YES	NULL	Reference number
invoice_id	UUID	YES	NULL	FK -> acc_invoices.id
expense_id	UUID	YES	NULL	FK -> acc_expenses.id
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 74: acc_tax_rates

Tax rate definitions.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Tax name (GST 18%, VAT 20%)
rate	DECIMAL(5,2)	NO	—	Rate
type	VARCHAR(50)	NO	'percentage'	percentage fixed
is_active	BOOLEAN	NO	true	Active status
created_at	TIMESTAMP	NO	NOW()	Creation time

MODULE 10 — INVENTORY (Tables 75-84)

TABLE 75: inv_categories

Product category tree.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Category name
parent_id	UUID	YES	NULL	FK -> inv_categories.id
description	TEXT	YES	NULL	Description
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 76: inv_units

Units of measure.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(100)	NO	—	Unit name (pieces, kg, liters)
abbreviation	VARCHAR(20)	NO	—	Short form (pcs, kg, L)
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 77: inv_products

Product and service catalog.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
sku	VARCHAR(100)	NO	—	UNIQUE stock keeping unit
name	VARCHAR(255)	NO	—	Product name
description	TEXT	YES	NULL	Description
category_id	UUID	YES	NULL	FK -> inv_categories.id
unit_id	UUID	YES	NULL	FK -> inv_units.id

Column	Type	Nullable	Default	Description
type	VARCHAR(50)	NO	'product'	product service
selling_price	DECIMAL(15,2)	YES	NULL	Sale price
cost_price	DECIMAL(15,2)	YES	NULL	Cost price
tax_rate	DECIMAL(5,2)	NO	0	Default tax %
image_url	TEXT	YES	NULL	Product image
barcode	VARCHAR(100)	YES	NULL	Barcode/QR code
is_active	BOOLEAN	NO	true	Active status
track_inventory	BOOLEAN	NO	true	Track stock
reorder_point	DECIMAL(10,2)	YES	NULL	Low stock threshold
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 78: inv_warehouses

Warehouse/location definitions.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Warehouse name
address	TEXT	YES	NULL	Address
is_primary	BOOLEAN	NO	false	Primary warehouse
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 79: inv_stock

Current stock levels per product per warehouse.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
product_id	UUID	NO	—	FK -> inv_products.id
warehouse_id	UUID	NO	—	FK -> inv_warehouses.id
quantity	DECIMAL(10,2)	NO	0	Available quantity
reserved_quantity	DECIMAL(10,2)	NO	0	Reserved for orders

Column	Type	Nullable	Default	Description
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 80: inv_stock_movements

All stock in/out/transfer events.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
product_id	UUID	NO	—	FK -> inv_products.id
warehouse_id	UUID	NO	—	FK -> inv_warehouses.id
type	VARCHAR(50)	NO	—	in out transfer adjustment
quantity	DECIMAL(10,2)	NO	—	Quantity moved
reference_type	VARCHAR(50)	YES	NULL	purchase_order sales_order
reference_id	UUID	YES	NULL	Reference record ID
notes	TEXT	YES	NULL	Notes
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 81: inv_purchase_orders

Supplier purchase orders.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
po_number	VARCHAR(100)	NO	—	PO number
vendor_id	UUID	NO	—	FK -> acc_vendors.id
warehouse_id	UUID	NO	—	FK -> inv_warehouses.id
status	VARCHAR(50)	NO	'draft'	draft sent received cancelled
order_date	DATE	NO	—	Order date
expected_date	DATE	YES	NULL	Expected delivery
received_date	DATE	YES	NULL	Actual receipt
subtotal	DECIMAL(15,2)	NO	0	Subtotal

Column	Type	Nullable	Default	Description
tax_amount	DECIMAL(15,2)	NO	0	Tax
total_amount	DECIMAL(15,2)	NO	0	Total
notes	TEXT	YES	NULL	Notes
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 82: inv_purchase_order_items

Line items within a purchase order.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
po_id	UUID	NO	—	FK -> inv_purchase_orders.id
product_id	UUID	NO	—	FK -> inv_products.id
quantity	DECIMAL(10,2)	NO	—	Ordered quantity
received_quantity	DECIMAL(10,2)	NO	0	Received quantity
unit_price	DECIMAL(15,2)	NO	—	Unit cost
tax_rate	DECIMAL(5,2)	NO	0	Tax rate %
amount	DECIMAL(15,2)	NO	—	Line total
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 83: inv_sales_orders

Customer sales orders.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
so_number	VARCHAR(100)	NO	—	SO number
customer_id	UUID	NO	—	FK -> acc_customers.id
warehouse_id	UUID	NO	—	FK -> inv_warehouses.id
status	VARCHAR(50)	NO	'draft'	draft confirmed shipped delivered cancelled

Column	Type	Nullable	Default	Description
order_date	DATE	NO	—	Order date
shipping_date	DATE	YES	NULL	Ship date
delivery_date	DATE	YES	NULL	Delivery date
subtotal	DECIMAL(15,2)	NO	0	Subtotal
tax_amount	DECIMAL(15,2)	NO	0	Tax
shipping_amount	DECIMAL(15,2)	NO	0	Shipping
total_amount	DECIMAL(15,2)	NO	0	Total
shipping_address	TEXT	YES	NULL	Ship-to address
notes	TEXT	YES	NULL	Notes
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 84: inv_sales_order_items

Line items within a sales order.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
so_id	UUID	NO	—	FK -> inv_sales_orders.id
product_id	UUID	NO	—	FK -> inv_products.id
quantity	DECIMAL(10,2)	NO	—	Ordered qty
shipped_quantity	DECIMAL(10,2)	NO	0	Shipped qty
unit_price	DECIMAL(15,2)	NO	—	Unit price
tax_rate	DECIMAL(5,2)	NO	0	Tax rate %
discount_percent	DECIMAL(5,2)	NO	0	Discount %
amount	DECIMAL(15,2)	NO	—	Line total
created_at	TIMESTAMP	NO	NOW()	Creation time

MODULE 11 — DOCUMENTS (Tables 85-88)

TABLE 85: doc_folders

Folder hierarchy for document storage.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	Folder name
parent_id	UUID	YES	NULL	FK -> doc_folders.id
created_by	UUID	NO	—	FK -> users.id
is_shared	BOOLEAN	NO	false	Shared flag
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 86: doc_files

Uploaded file records (stored in MinIO).

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
folder_id	UUID	YES	NULL	FK -> doc_folders.id
name	VARCHAR(255)	NO	—	File name
description	TEXT	YES	NULL	Description
url	TEXT	NO	—	MinIO storage URL
size	BIGINT	NO	—	File size bytes
mime_type	VARCHAR(100)	NO	—	MIME type
extension	VARCHAR(20)	YES	NULL	File extension
version	INTEGER	NO	1	Current version
is_public	BOOLEAN	NO	false	Public access
public_token	VARCHAR(100)	YES	NULL	UNIQUE share token
tags	TEXT[]	NO	'{}'	Tags

Column	Type	Nullable	Default	Description
uploaded_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 87: doc_file_versions

Version history for files.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
file_id	UUID	NO	—	FK -> doc_files.id
version	INTEGER	NO	—	Version number
url	TEXT	NO	—	Version file URL
size	BIGINT	NO	—	File size
change_summary	TEXT	YES	NULL	What changed
uploaded_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Upload time

TABLE 88: doc_shares

File and folder sharing permissions.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
file_id	UUID	YES	NULL	FK -> doc_files.id
folder_id	UUID	YES	NULL	FK -> doc_folders.id
shared_with	UUID	NO	—	FK -> users.id
permission	VARCHAR(20)	NO	'view'	view edit admin
shared_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time

MODULE 12 — TEAM CHAT (Tables 89-93)

TABLE 89: chat_channels

Public, private, and DM channels.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(100)	NO	—	Channel name
description	TEXT	YES	NULL	Description
type	VARCHAR(20)	NO	'public'	public private direct
created_by	UUID	NO	—	FK -> users.id
is_archived	BOOLEAN	NO	false	Archived flag
archived_at	TIMESTAMP	YES	NULL	Archive time
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

TABLE 90: chat_channel_members

Users in a channel.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
channel_id	UUID	NO	—	FK -> chat_channels.id
user_id	UUID	NO	—	FK -> users.id
role	VARCHAR(20)	NO	'member'	owner admin member
last_read_at	TIMESTAMP	YES	NULL	Last read time
is_muted	BOOLEAN	NO	false	Muted flag
joined_at	TIMESTAMP	NO	NOW()	Join time

TABLE 91: chat_messages

Individual chat messages.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key

Column	Type	Nullable	Default	Description
channel_id	UUID	NO	—	FK -> chat_channels.id
user_id	UUID	NO	—	FK -> users.id
content	TEXT	YES	NULL	Message text
type	VARCHAR(50)	NO	'text'	text file image system
parent_id	UUID	YES	NULL	FK -> chat_messages.id (thread)
reply_count	INTEGER	NO	0	Thread replies
is_edited	BOOLEAN	NO	false	Edited flag
edited_at	TIMESTAMP	YES	NULL	Edit time
created_at	TIMESTAMP	NO	NOW()	Send time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 92: chat_message_attachments

Files sent in chat.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
message_id	UUID	NO	—	FK -> chat_messages.id
name	VARCHAR(255)	NO	—	File name
url	TEXT	NO	—	File URL
size	BIGINT	YES	NULL	File size
mime_type	VARCHAR(100)	YES	NULL	MIME type
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 93: chat_reactions

Emoji reactions on messages.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
message_id	UUID	NO	—	FK -> chat_messages.id
user_id	UUID	NO	—	FK -> users.id
emoji	VARCHAR(50)	NO	—	Emoji code

Column	Type	Nullable	Default	Description
created_at	TIMESTAMP	NO	NOW()	Creation time

MODULE 13 — APP BUILDER (Tables 94-98)

TABLE 94: builder_apps

Low-code app definitions.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
organization_id	UUID	NO	—	FK -> organizations.id
name	VARCHAR(255)	NO	—	App name
description	TEXT	YES	NULL	Description
slug	VARCHAR(100)	NO	—	URL identifier
icon	VARCHAR(50)	YES	NULL	Lucide icon name
color	VARCHAR(20)	NO	'#6366f1'	Theme color
status	VARCHAR(50)	NO	'draft'	draft published
settings	JSONB	NO	'{}'	App settings
created_by	UUID	NO	—	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 95: builder_app_tables

Custom data tables within an app.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
app_id	UUID	NO	—	FK -> builder_apps.id
name	VARCHAR(255)	NO	—	Table display name
slug	VARCHAR(100)	NO	—	Table identifier
description	TEXT	YES	NULL	Description
icon	VARCHAR(50)	YES	NULL	Table icon
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 96: builder_app_fields

Column definitions for app builder tables.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
table_id	UUID	NO	—	FK -> builder_app_tables.id
name	VARCHAR(255)	NO	—	Field display name
slug	VARCHAR(100)	NO	—	Field identifier
type	VARCHAR(50)	NO	—	text number date boolean select multi_select file relation formula url email phone
options	JSONB	YES	NULL	Select options array
is_required	BOOLEAN	NO	false	Required flag
is_unique	BOOLEAN	NO	false	Unique constraint
default_value	TEXT	YES	NULL	Default value
formula	TEXT	YES	NULL	Formula expression
order	INTEGER	NO	0	Display order
created_at	TIMESTAMP	NO	NOW()	Creation time

TABLE 97: builder_app_records

Data rows in app builder tables.

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
table_id	UUID	NO	—	FK -> builder_app_tables.id
data	JSONB	NO	'{}'	All field values as JSON
created_by	UUID	YES	NULL	FK -> users.id
updated_by	UUID	YES	NULL	FK -> users.id
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time
deleted_at	TIMESTAMP	YES	NULL	Soft delete

TABLE 98: builder_app_views

Page views within an app (list, form, kanban...).

Column	Type	Nullable	Default	Description
id	UUID	NO	gen_random_uuid()	Primary key
app_id	UUID	NO	—	FK -> builder_apps.id
name	VARCHAR(255)	NO	—	View name
type	VARCHAR(50)	NO	—	list form detail calendar kanban gallery map
table_id	UUID	YES	NULL	FK -> builder_app_tables.id
config	JSONB	NO	'{}'	Layout, columns, filters config
order	INTEGER	NO	0	Navigation order
is_home	BOOLEAN	NO	false	Default/home view
created_at	TIMESTAMP	NO	NOW()	Creation time
updated_at	TIMESTAMP	NO	NOW()	Last update time

9. Installation Guide

Step 1	npm install -g pnpm
Step 2	git init (inside project folder)
Step 3	docker compose up -d
Step 4	pnpm install
Step 5	pnpm db:push (apply Prisma schema)
Step 6	pnpm db:seed (seed initial data)
Step 7	pnpm dev (start dev server on port 3000)

10. Final Summary

Total Tables	98
Total Columns	~1,090
Total Modules	13
Total Files (Full)	~1,200+
Total Files (Phase 1)	~180
Backend API Procedures	~200+ (tRPC)
React Components	~300+
Pages / Routes	~60+
Docker Services	5 (PostgreSQL, Redis, MinIO, Mailhog, MinIO Console)
Auth Methods	Email, Google OAuth, Microsoft OAuth, GitHub OAuth, MFA/TOTP
Build Duration (Estimate)	~52 weeks (13 phases x 4 weeks)
Infrastructure Cost (Local)	\$0 — all Docker + free-tier services

Generated: 2026-05-27 | SaaS Platform v1.0 | Author: Mohit Yadav | Mobilise App Lab